



NTU Academy for Professional
and Continuing Education

(SCTP) Advanced
Professional Certificate

Data Science and AI



2.1 Introduction to Big Data and Data Engineering

Module Overview

2.1 Introduction to Big Data and Data Engineering

2.2 Data Architecture

2.3 Data Encoding and Data Flow

2.4 Data Extraction and Web Scraping

2.5 Data Warehouse

2.6 Data Pipelines and Orchestration

2.7 Data Orchestration and Testing

2.8 Out of Core/Memory Processing

2.9 Big Data Ecosystem and Batch Processing

2.10 Event Streaming and Stream Processing

Agenda

- Introduction to Big Data, 5Vs
- Big Data Tools and Technologies
- Structured vs Unstructured Data
- NoSQL Database Types
- OLTP vs OLAP
- ACID vs BASE
- Introduction to Data Engineering
- Data Engineering Lifecycle

Big Data

- Volume and Variety
 - Massive amounts of structured and unstructured data
- Velocity and Veracity
 - Fast moving data streams requiring reliable processing

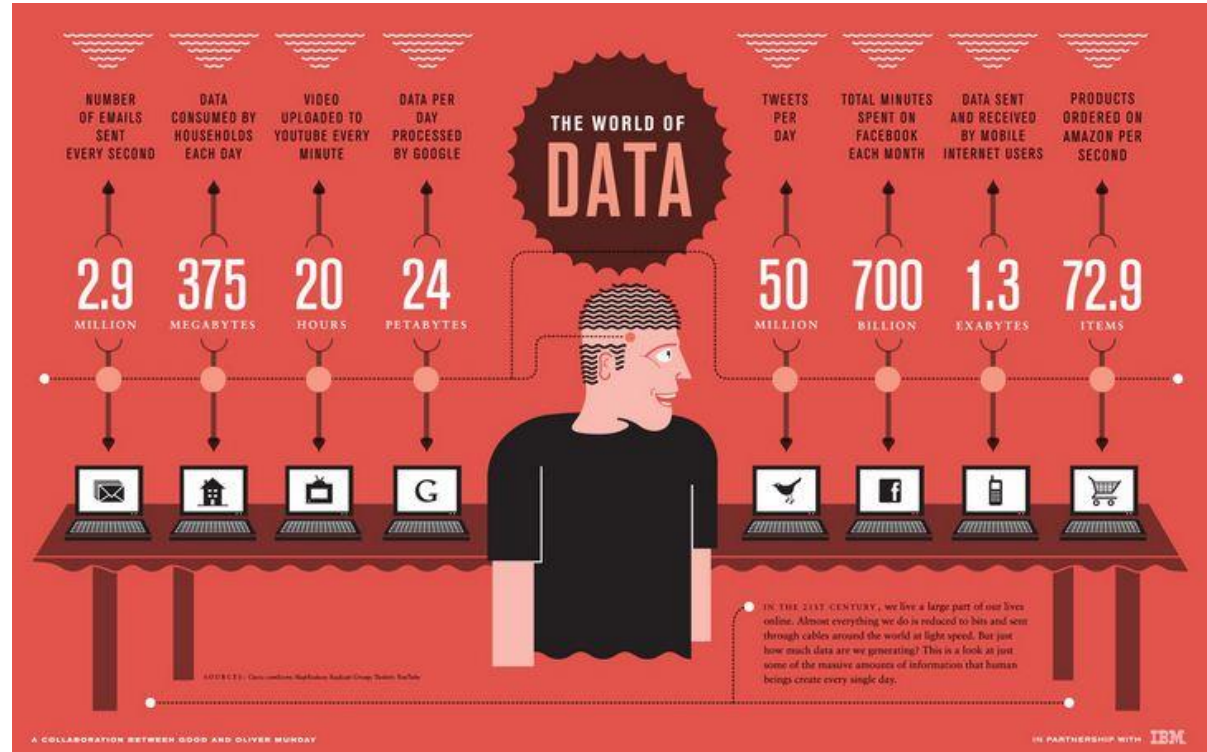
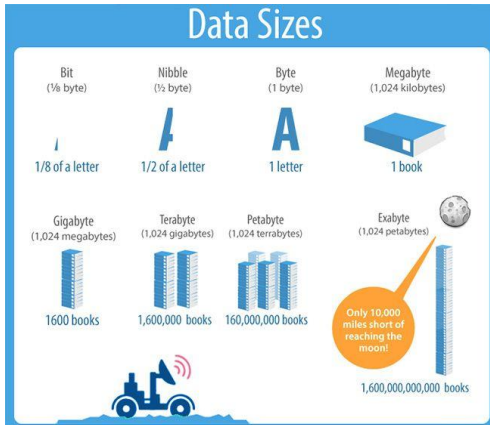
Data Engineering

- Lifecycle management
 - Generation
 - Storage
 - Ingestion
 - Transform
 - Serve
- Data engineering provides the framework to extract value from big data

Big Data describes large volume of data:

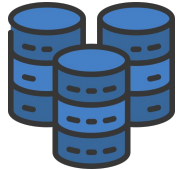
- *structured* and *unstructured*

Large amounts of data that cannot be processed using traditional methods.

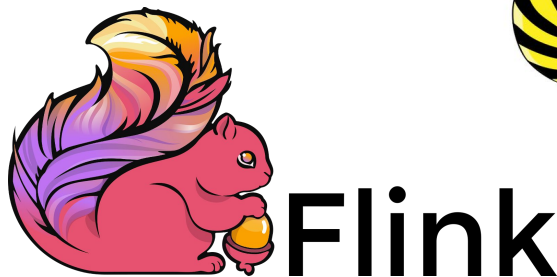
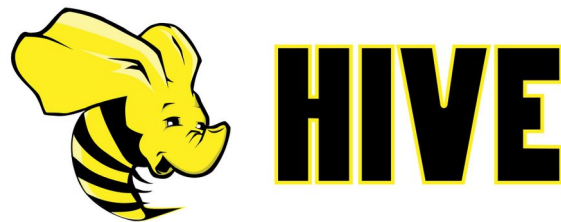


5 Vs of Big Data

- **Volume:** The total amount of data produced by both machines and people.
- **Velocity:** How fast data is created and processed.
- **Variety:** The range of data types, encompassing both structured and unstructured data.
- **Veracity:** The reliability of data, including its biases, noise, and abnormalities, affecting insight accuracy.
- **Value:** The usefulness of data, defined by the insights it enables, the decisions it guides, and its impact on business outcomes.

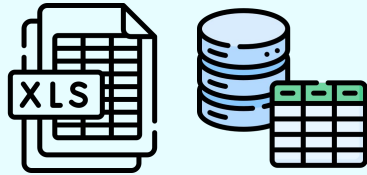


Big Data Tools and Technologies



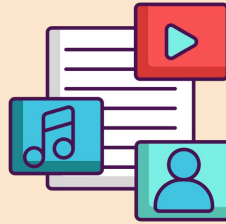
Structured and Unstructured Data

Structured Data



Organized in tables, easy to search/analyze. *Examples: Relational databases (DuckDB), spreadsheets (Excel).*

Unstructured Data



Lacks predefined organization, hard to search/analyze. *Examples: images, videos, social media posts. Makes up ~80% of generated data.*

Relational databases need structured schema

NoSQL databases are better for storing large amount of unstructured data, offering scalability and flexibility.

Database Recap



Relational Database

- Also known as SQL database
- Based on relational model and has predefined schema
- Suited for structured data
- Uses SQL (Structured Query Language) for querying



NoSQL Database

- “Not-only” SQL, or non-relational database
- Flexible schema
- Suited for semi-structured / unstructured data
- SQL optional

OLTP vs. OLAP: The Engine vs. The Brain of Data

Modern data engineering relies on two distinct systems: **OLTP** for managing day-to-day business operations and **OLAP** for analyzing those operations to gain insights. Understanding when to use each is fundamental to building efficient data architectures.

OLTP (Online Transaction Processing)

The “Engine” of Daily Operations

Why? To record and process every business interaction instantly and accurately.



Real-World Scenario: ATM Withdrawals

When you withdraw cash, the system must immediately update your balance to prevent overspending.

High-Speed Read/Write

Optimized for fast, frequent updates to small amounts of data (normalized).

OLAP (Online Analytical Processing)

The “Brain” of Business Strategy

Why? To identify long-term trends and patterns for better decision-making.



Real-World Scenario: Annual Sales Trends

A retail manager analyzes three years of sales data to predict next year's inventory.

High-Volume Read-Only

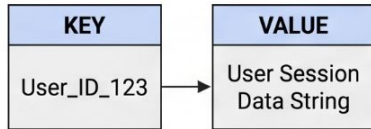
Optimized for complex queries involving massive datasets and aggregations (denormalized).

Comparison Summary

Feature	OLTP (Operations)	OLAP (Analysis)
Primary Goal	Process Transactions	Analyze Data
Data Operation	Read and Write (Frequent)	Mostly Read (Complex Queries)
Data State	Normalized (Saves Space)	Denormalized (Fast Queries)

Types of NoSQL databases

Key-Value Stores



Optimized for High-Speed Lookups & Horizontal Scaling

Perfect for Session Management and Real-World E-commerce Carts



Redis



DynamoDB

Column-Family Databases

	Sensor_Core		Environmental_Data	
	Temp	Humidity	Temp	Humidity
keys	33.5	24	25	64
...				
key1				
key2				
key3				
key4				

Efficient for Querying Specific Column Subsets in Massive Datasets

Ideal for Analyzing Large IoT Telemetries with Varying Attributes

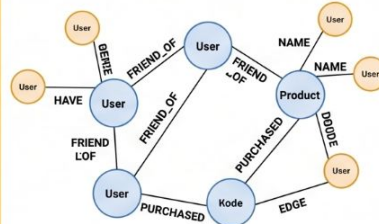


Cassandra



HBase

Graph Databases



Specialized for Querying Complex Relationships and Discovering Patterns

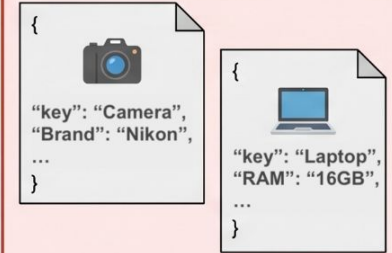
Grounded in Social Networks and Fraud Detection Scenarios



Neo4j



Document Stores



Flexible Schemas for Diverse & Semi-Structured Data

Suited for Content Management and Complex Product Catalogues



MongoDB



CouchDB

ACID

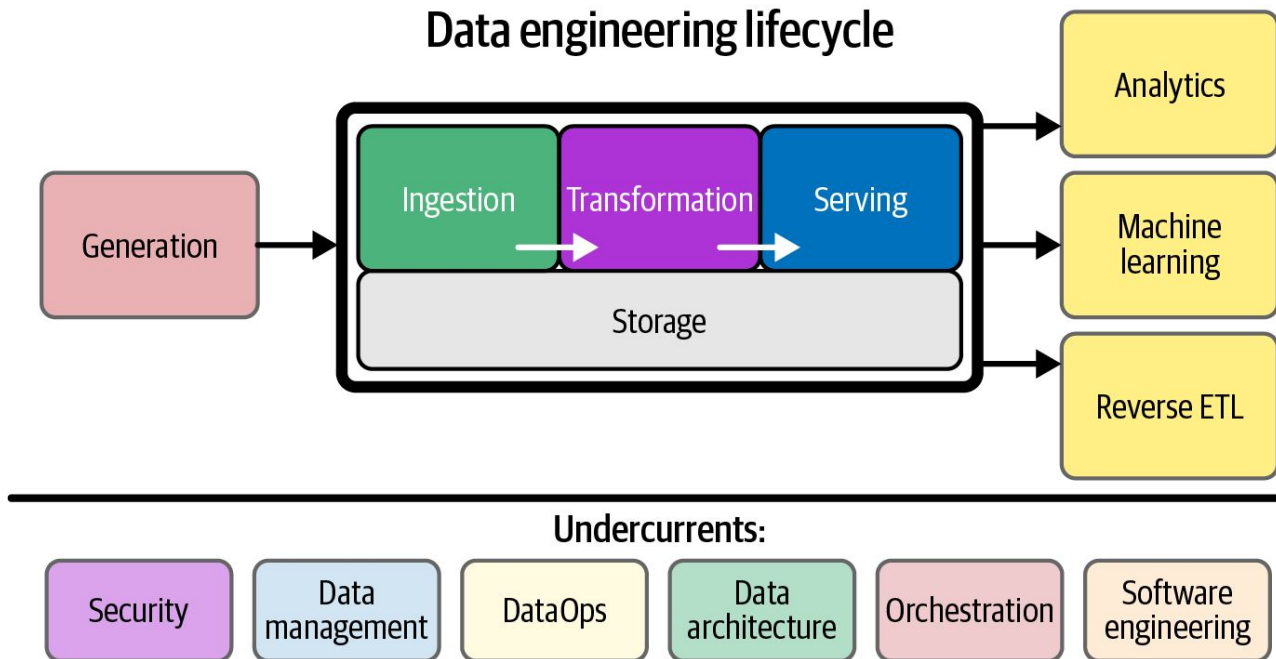
- **A**tomicity, **C**onsistency, **I**solation, **D**urability
- Guarantees all transactions are valid, even in event of failures
- High data integrity and reliability
- **Can lead to performance bottlenecks**
- **Challenging to scale**

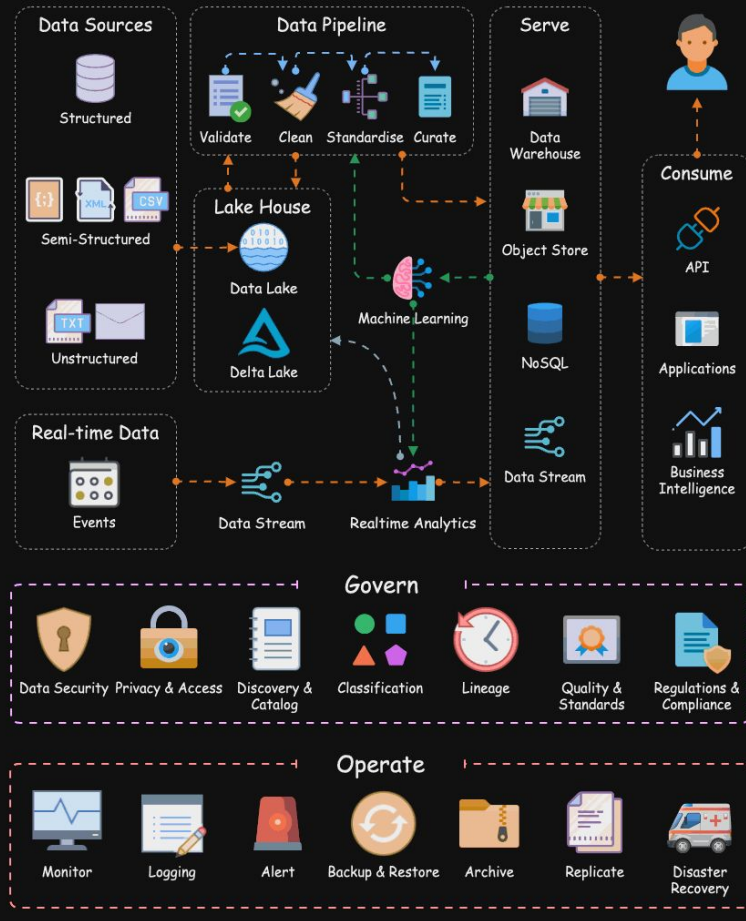
BASE

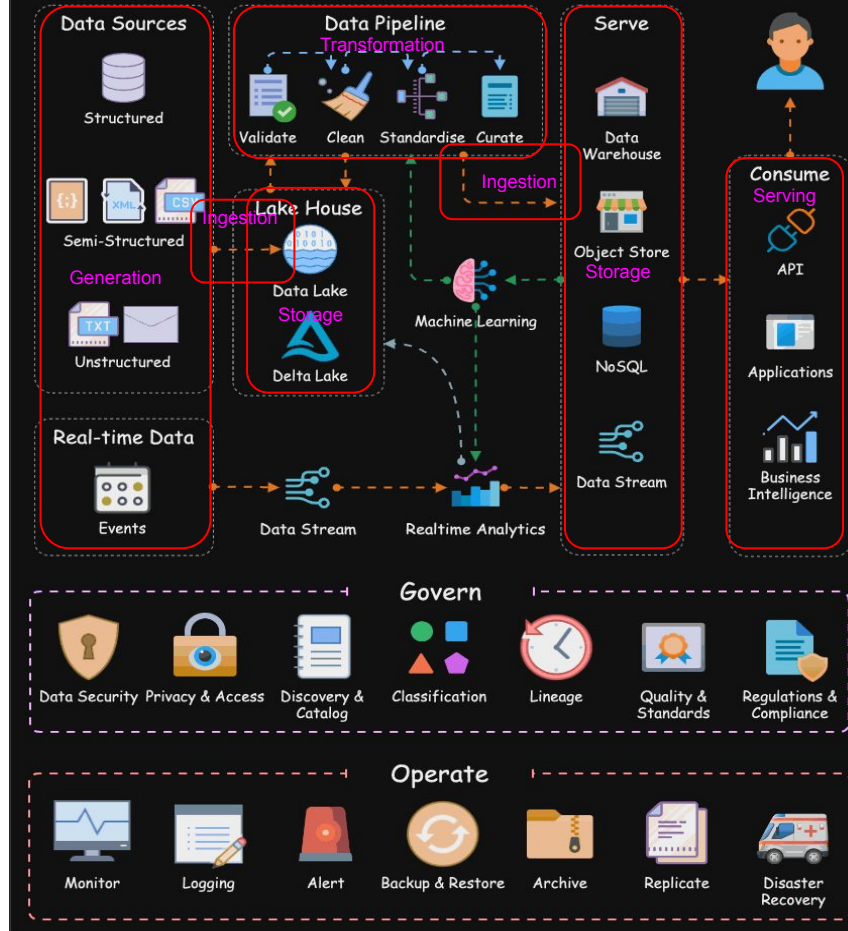
- **B**asically **A**vailable, **S**oft state, **E**ventual consistency
- Prioritize availability and scalability over immediate consistency
- Lower latency and higher throughput
- Relaxes some ACID properties
- **Temporary data inconsistency**
- **Possible stale data**

Data Engineering Lifecycle

- Generation
- Storage
- Ingestion
- Transformation
- Serving







End of Lesson - Exit Ticket

Slido Survey Link will be provided in class

